

The numerical range is a 2-spectral set

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Abstract

For an $n \times n$ complex matrix A , let $W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}$ be its numerical range. Here x^* denotes the conjugate transpose, $\|\cdot\|_2$ the Euclidean norm, and $\|\cdot\|$ the induced operator norm. For every complex polynomial p , we prove that

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|$$

and consequently that $W(A)$ is a 2-spectral set for A . This proves Crouzeix's conjecture, and the constant is optimal. The central new ingredient is a positive-real completion theorem relative to an auxiliary eigenbasis: an adjoint-algebra constraint on the resolvent defect of a normalized matrix-valued Carathéodory function forces the underlying matrix to have norm at most 2. The classical positive double-layer calculus supplies such completions for $f(B)$ whenever the auxiliary matrix B has simple spectrum; the eigenvalues of $f(B)$ need not be distinct. Sampling the associated Herglotz kernel at half the conjugate diagonal entries and at the origin cancels the completion term and leaves a comparison of two weighted Gramians; a Stein identity yields the sharp estimate. Simple-spectrum approximation and smooth convex outer approximation remove the remaining auxiliary hypotheses.

Keywords: Crouzeix conjecture; numerical range; spectral set; matrix-valued Herglotz kernel; double-layer potential

MSC: Primary 47A25; Secondary 47A12, 15A60

1. Introduction

Let $\mathbb{C}$ denote the complex field and, for $n \geq 1$, let $\mathbb{M}_n(\mathbb{C})$ denote the algebra of $n \times n$ complex matrices. For $A \in \mathbb{M}_n(\mathbb{C})$, its numerical range is

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}. \quad (1)$$

Here x^* denotes the conjugate transpose, $\sigma(A)$ denotes the spectrum, $\|x\|_2$ is the Euclidean norm, $\|X\|$ is the induced operator norm, and scalar moduli are denoted by $|\cdot|$. The Toeplitz–Hausdorff theorem shows that $W(A)$ is compact and convex, and $\sigma(A) \subset W(A)$.

A compact set $K \subset \mathbb{C}$ containing $\sigma(A)$ is called a C -spectral set for A if

$$\|r(A)\| \leq C \max_{z \in K} |r(z)| \quad (2)$$

for every rational function r with no pole on K . Our main result settles Crouzeix's conjecture by identifying the sharp universal constant for $K = W(A)$.

Theorem 1.1 (Main theorem). *Let $n \geq 1$, let $A \in \mathbb{M}_n(\mathbb{C})$, and let $p \in \mathbb{C}[z]$. Then*

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|. \quad (3)$$

Consequently, $W(A)$ is a 2-spectral set for A .

The factor 2 is optimal: for

$$N = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad p(z) = z, \quad (4)$$

one has $W(N) = \{z \in \mathbb{C} : |z| \leq 1\}$ and $\|N\| = 2$.

For later use, write

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}, \quad \mathbb{T} = \partial\mathbb{D}.$$

We use I for the identity matrix of the size dictated by context, X^* for the adjoint, put $\operatorname{Re} X = (X + X^*)/2$, and let $\operatorname{alg}(T)$ denote the unital algebra generated by T . For Hermitian matrices X and Y , we write $X \preceq Y$ (equivalently, $Y \succeq X$) when $Y - X$ is positive semidefinite.

Crouzeix proved the estimate with constant 2 for 2×2 matrices and formulated the general conjecture in [1]. His subsequent dimension-free theorem gave the constant 11.08 [2]. After further structural and numerical work [3], Crouzeix and Palencia proved that $W(A)$ is a $(1 + \sqrt{2})$ -spectral set [4]. Their double-layer and Cauchy-transform argument was clarified in [5] and developed in several directions in [6,7].

The intervening literature established the conjectured constant for perturbed Jordan blocks and, more generally, scalar translates of cyclic weighted shifts, as well as for tridiagonal 3×3 matrices with elliptic numerical range centered at an eigenvalue [8–10]. Numerical and extremal investigations appear in [11,12], and a broad account of the problem and its variants is given in [13]. Subsequent work has treated Blaschke-product level sets, cyclicity reductions, compressions of shifts, further nilpotent classes, and quantitative bounds for KMS matrices [14–17]. An alternative proof for a class of weighted shifts appears in [18]. None of the cited reductions supplies a dimension-free argument that lowers the universal bound from $1 + \sqrt{2}$ to 2.

1.1. Where the earlier mechanism loses sharpness

The positive boundary measure used below originates in the convex-domain integral calculus of Delyon–Delyon [19]; see also [4,20]. If B has numerical range in a smooth convex domain Ω and f is bounded by one on Ω , this calculus produces a unital positive map Φ and a holomorphic Cauchy companion g_f such that

$$2\Phi(f) = f(B) + g_f(B)^*. \quad (5)$$

Thus positivity directly controls a symmetrized functional calculus, not $f(B)$ by itself. The Crouzeix–Palencia argument couples the two terms in (5), and its abstract norm estimate is sharp at $1 + \sqrt{2}$. The examples in [5] show that this abstract mechanism alone cannot give 2. The corresponding general uniform-algebra theorem is likewise sharp at $1 + \sqrt{2}$; its additional unital case was proposed as a possible route to 2 in [21]. Schwenninger and de Vries revisited the double-layer method [22]. Malman et al. proved that a configuration-constant refinement gives a strict improvement for every fixed numerical-range shape; they also constructed thin quadrilaterals for which the configuration constant approaches its limiting value, so it yields no smaller universal constant [23]. Separately, continuity and compactness give a constant $C_n < 1 + \sqrt{2}$ in every fixed matrix dimension n [24].

The missing information is algebraic. The companion in (5) is not an arbitrary operator: for the approximants constructed in Section 5, it belongs to the algebra generated by B^* , where the simple-spectrum matrix B supplies an eigenbasis that also diagonalizes $f(B)$. Estimating the companion as an unrelated summand discards precisely this coupling.

1.2. The new de-symmetrization mechanism

Our argument retains the whole Cayley family, rather than the single identity (5). It has four steps.

1. Applying Φ to

$$\zeta \mapsto \frac{1 + wf(\zeta)}{1 - wf(\zeta)}, \quad w \in \mathbb{D},$$

produces a matrix-valued Carathéodory function H , meaning that $H(0) = I$ and $\operatorname{Re} H(w) \succeq 0$.

2. For a simple-spectrum auxiliary matrix B , put $T = f(B)$. The double-layer identity gives the positive-real completion

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*), \quad w \in \mathbb{D}. \quad (6)$$

3. The eigenbasis of B diagonalizes $T = f(B)$ even when some values of f on the spectrum of B coincide, and it turns the unknown term in (6) into a diagonal analytic correction. We sample the Herglotz kernel of H at half the conjugate diagonal entries of T and add one vector sample at the origin. Repeated sampling points are allowed, and the origin sample cancels the diagonal correction exactly.

4. The surviving positive-matrix inequality compares two weighted Gramians. Its difference is positive, and an eigenvector argument followed by a Stein identity yields $\|T\| \leq 2$.

No perturbation of f is required. We first let the simple-spectrum matrices tend to A and then let the outer domains converge to $W(A)$. These two limits remove the remaining auxiliary hypotheses.

2. Positive-real kernels

Let $\mathcal{K}$ be an $\mathbb{M}_n(\mathbb{C})$ -valued kernel on $\mathbb{D} \times \mathbb{D}$. We call $\mathcal{K}$ positive if, for every finite choice $w_0, \dots, w_m \in \mathbb{D}$ and $\xi_0, \dots, \xi_m \in \mathbb{C}^n$,

$$\sum_{i,j=0}^m \xi_i^* \mathcal{K}(w_i, w_j) \xi_j \geq 0. \quad (7)$$

We shall use the following standard matrix-valued form of the Herglotz theorem. Its kernel consequence is included to fix the normalization.

Lemma 2.1 (Matrix Herglotz kernel). *Let $F: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic with $\operatorname{Re} F(w) \succeq 0$ for every $w \in \mathbb{D}$. Then*

$$\mathcal{L}_F(w, z) = \frac{F(w) + F(z)^*}{1 - w\bar{z}} \quad (8)$$

is a positive kernel.

Proof. The matrix Herglotz representation gives a positive $\mathbb{M}_n(\mathbb{C})$ -valued measure M on $\mathbb{T}$, and a Hermitian matrix J_F , such that

$$F(w) = iJ_F + \int_{\mathbb{T}} \frac{\zeta + w}{\zeta - w} dM(\zeta). \quad (9)$$

Indeed, for each $u \in \mathbb{C}^n$ the scalar Herglotz theorem gives a positive finite measure μ_u on $\mathbb{T}$ and a real number c_u such that

$$u^* F(w) u = ic_u + \int_{\mathbb{T}} \frac{\zeta + w}{\zeta - w} d\mu_u(\zeta).$$

Uniqueness in the scalar theorem shows that, for every Borel set $E \subset \mathbb{T}$, the map $u \mapsto \mu_u(E)$ is a nonnegative quadratic form. Indeed, uniqueness applied to the corresponding identities for the scalar functions gives, as identities of measures,

$$\mu_{\alpha u} = |\alpha|^2 \mu_u, \quad \mu_{u+v} + \mu_{u-v} = 2\mu_u + 2\mu_v \quad (\alpha \in \mathbb{C}, u, v \in \mathbb{C}^n).$$

Its polarization defines

$$x^* M(E) y = \frac{1}{4} \sum_{k=0}^3 i^{-k} \mu_{x+i^k y}(E) \quad (x, y \in \mathbb{C}^n).$$

Thus M is a finite $\mathbb{M}_n(\mathbb{C})$ -valued measure and $u^* M(E) u = \mu_u(E) \geq 0$, so M is positive. The scalar imaginary constants polarize to the Hermitian matrix $J_F = (F(0) - F(0)^*)/(2i)$, and the scalar representations assemble to (9).

For $|\zeta| = 1$, direct calculation gives

$$\frac{1}{1 - w\bar{z}} \left(\frac{\zeta + w}{\bar{\zeta} - \bar{w}} + \frac{\bar{\zeta} + z}{\zeta - z} \right) = \frac{2}{(1 - w\bar{\zeta})(1 - \bar{z}\zeta)}. \quad (10)$$

The constant iJ_F cancels from (8). Substituting (9) and then summing as in (7) yields

$$2 \int_{\mathbb{T}} \left(\sum_i \frac{\xi_i}{1 - \bar{w}_i \zeta} \right)^* dM(\zeta) \left(\sum_j \frac{\xi_j}{1 - \bar{w}_j \zeta} \right) \geq 0. \quad \square$$

3. The positive-real completion principle

The following completion theorem is the core of the argument: it converts a positive-real completion constrained by the adjoint algebra of an auxiliary simple-spectrum matrix into the sharp norm bound.

Theorem 3.1 (Positive-real completion). *Let*

$$B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}, \quad T = S \Lambda S^{-1}, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \quad (11)$$

where S is invertible, the numbers $\beta_1, \dots, \beta_n$ are distinct, and $|\lambda_i| \leq 1$ for every i . The numbers λ_i are not required to be distinct. Let $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic and satisfy

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (12)$$

and

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (13)$$

Then $\|T\| \leq 2$.

Proof. Use the simultaneous diagonalizations in (11) and put $G = S^* S$. Since the β_i are distinct, polynomial interpolation shows that

$$\operatorname{alg}(B^*) = \left\{ (S^{-1})^* D S^* : D \text{ is diagonal} \right\}. \quad (14)$$

Set

$$\Delta(w) = H(w) - (I - wT)^{-1}.$$

By (14), for each $w \in \mathbb{D}$ there is a unique diagonal matrix $\Theta(w)$ such that

$$\Delta(w) = (S^{-1})^* \Theta(w) S^*.$$

In fact,

$$\Theta(w) = S^* \Delta(w) (S^*)^{-1},$$

so Θ is analytic; moreover, $\Theta(0) = 0$ because $H(0) = I$. Write $\Theta(w) = \text{diag}(\theta_1(w), \dots, \theta_n(w))$. On setting $\tilde{H}(w) = S^*H(w)S$, we obtain

$$\tilde{H}(w) = G(I - w\Lambda)^{-1} + \Theta(w)G. \quad (15)$$

Since $\text{Re } \tilde{H}(w) = S^*(\text{Re } H(w))S$, the real part of $\tilde{H}(w)$ is positive semidefinite. Consequently, Lemma 2.1 applies to $\tilde{H}$.

Define Hermitian matrices $P, Q, Y \in \mathbb{M}_n(\mathbb{C})$ by

$$P_{ij} = \frac{G_{ij}}{1 - \bar{\lambda}_i \lambda_j / 4}, \quad Q_{ij} = \frac{G_{ij}}{1 - \bar{\lambda}_i \lambda_j / 2}, \quad Y = Q - P. \quad (16)$$

Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{C}^n$, and fix $u = (u_1, \dots, u_n)^T \in \mathbb{C}^n$. In the positive-kernel inequality use the n points and vectors

$$w_i = \frac{\bar{\lambda}_i}{2}, \quad \xi_i = u_i e_i \quad (1 \leq i \leq n), \quad (17)$$

and add the point $w_0 = 0$ with vector

$$\xi_0 = v = -G^{-1}Pu. \quad (18)$$

The points w_i need not be distinct. Kernel positivity applies to arbitrary finite lists of points and vectors, so repetitions among the w_i , including $w_i = w_0$ when $\lambda_i = 0$, cause no difficulty. We first verify that this sample cancels the entire unknown function Θ . The contribution of the second term in (15) to the kernel quadratic form is

$$2 \text{Re} \sum_{i=1}^n \bar{u}_i \theta_i(w_i) \left((Gv)_i + \sum_{j=1}^n \frac{G_{ij} u_j}{1 - w_i \bar{w}_j} \right). \quad (19)$$

Because

$$1 - w_i \bar{w}_j = 1 - \bar{\lambda}_i \lambda_j / 4,$$

the parenthesis in (19) is $(Gv + Pu)_i$, which vanishes by (18).

It remains to calculate the contribution of $G(I - w\Lambda)^{-1}$. Between w_i and w_j its (i, j) entry is

$$\frac{2G_{ij}}{(1 - \bar{\lambda}_i \lambda_j / 2)(1 - \bar{\lambda}_i \lambda_j / 4)} = (4Q - 2P)_{ij}. \quad (20)$$

Between the eigenvalue samples and the origin, the two blocks are $G + Q$, and the block at the origin is $2G$. After the cancellation of Θ , kernel positivity therefore gives

$$\begin{aligned} 0 &\leq \begin{pmatrix} u \\ v \end{pmatrix}^* \begin{pmatrix} 4Q - 2P & G + Q \\ G + Q & 2G \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} \\ &= u^* ((4Q - 2P) - (G + Q)G^{-1}P - PG^{-1}(G + Q) + 2PG^{-1}P)u \\ &= u^* (4Y - YG^{-1}P - PG^{-1}Y)u, \end{aligned} \quad (21)$$

where in the second line we used $v = -G^{-1}Pu$, and in the last line we used $Q = P + Y$. Since u was arbitrary,

$$4Y - YG^{-1}P - PG^{-1}Y \succeq 0. \quad (22)$$

Balance the diagonalization by defining

$$\begin{aligned}\tilde{T} &= G^{1/2}\Lambda G^{-1/2}, & \hat{P} &= G^{-1/2}PG^{-1/2}, \\ \hat{Q} &= G^{-1/2}QG^{-1/2}, & \hat{Y} &= \hat{Q} - \hat{P}.\end{aligned}\tag{23}$$

Expanding the scalar kernels in (16) into geometric series gives

$$\hat{P} = \sum_{k=0}^{\infty} 4^{-k} \tilde{T}^{*k} \tilde{T}^k, \quad \hat{Q} = \sum_{k=0}^{\infty} 2^{-k} \tilde{T}^{*k} \tilde{T}^k.\tag{24}$$

Indeed, $\tilde{T}^k = G^{1/2}\Lambda^k G^{-1/2}$, so the sequence $(\tilde{T}^k)$ is bounded even when some $|\lambda_j| = 1$; the geometric weights give norm convergence. It follows that

$$\hat{Y} = \sum_{k=1}^{\infty} (2^{-k} - 4^{-k}) \tilde{T}^{*k} \tilde{T}^k \succeq 0.\tag{25}$$

Taking the congruence of (22) by $G^{-1/2}$ yields

$$4\hat{Y} - \hat{Y}\hat{P} - \hat{P}\hat{Y} \succeq 0.\tag{26}$$

Let e be an eigenvector of the Hermitian matrix $\hat{P}$, say $\hat{P}e = \alpha e$. If $\alpha > 2$, then (26) gives

$$0 \leq 2(2 - \alpha)e^* \hat{Y} e.\tag{27}$$

Since $\hat{Y} \succeq 0$ and $\alpha > 2$, (27) forces $e^* \hat{Y} e = 0$. Using the series in (25), we obtain

$$0 = e^* \hat{Y} e = \sum_{k=1}^{\infty} (2^{-k} - 4^{-k}) \|\tilde{T}^k e\|_2^2 \geq \frac{1}{4} \|\tilde{T} e\|_2^2.$$

Thus $\tilde{T}e = 0$, and the first series in (24) gives $\hat{P}e = e$, contradicting $\hat{P}e = \alpha e$ with $\alpha > 2$. The first series in (24) also begins with I , and hence

$$I \preceq \hat{P} \preceq 2I.\tag{28}$$

Finally, the first Gramian satisfies the Stein identity

$$\hat{P} = I + \frac{1}{4} \tilde{T}^* \hat{P} \tilde{T}.\tag{29}$$

Using (28), we obtain

$$\tilde{T}^* \tilde{T} \preceq \tilde{T}^* \hat{P} \tilde{T} = 4(\hat{P} - I) \preceq 4I.$$

Thus $\|\tilde{T}\| \leq 2$. The polar decomposition $S = UG^{1/2}$ gives

$$T = U\tilde{T}U^*,$$

where U is unitary. Therefore $\|T\| \leq 2$. $\square$

Remark 3.2. The factor $1/2$ in the sampling points (17) simultaneously produces the weights $1/2$ and $1/4$ in (24). Their positive difference and the Stein identity are what recover the sharp constant. The origin sample removes the unknown analytic correction exactly; no estimate is used at this step. Distinctness is needed only for the eigenvalues of the auxiliary matrix B , which identify $\text{alg}(B^*)$ with the diagonal algebra in the chosen basis; no distinctness is required of the diagonal entries of T .

4. The double-layer completion

We now construct from the numerical range the function required by Theorem 3.1. The measure and companion transform used here are classical [2,4,19]. The new point is that retaining the full Cayley family yields the algebraic constraint (13) relative to the auxiliary matrix.

Lemma 4.1 (Double-layer positive map). *Let $B \in \mathbb{M}_n(\mathbb{C})$, and let $\Omega \subset \mathbb{C}$ be a bounded convex domain with continuously differentiable boundary and $W(B) \subset \Omega$. Writing $C(\partial\Omega)$ for the continuous complex-valued functions on $\partial\Omega$, there is a unital positive map*

$$\Phi: C(\partial\Omega) \longrightarrow \mathbb{M}_n(\mathbb{C})$$

such that, whenever h is holomorphic on a neighborhood of $\overline{\Omega}$,

$$2\Phi(h) = h(B) + g_h(B)^*, \quad (30)$$

where

$$g_h(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{h(\zeta)}}{\zeta - z} d\zeta, \quad z \in \Omega. \quad (31)$$

Proof. Orient $\partial\Omega$ counterclockwise. Let $\nu(\zeta)$ be the outward unit normal, let ds be arclength, and put

$$Z_\zeta = \zeta I - B, \quad \mathcal{R}_\zeta = Z_\zeta^{-1}.$$

This resolvent exists because $\sigma(B) \subset W(B) \subset \Omega$. Define the matrix-valued boundary measure

$$d\mu_B(\zeta) = \frac{1}{2\pi} \left(\nu(\zeta) \mathcal{R}_\zeta + \overline{\nu(\zeta)} \mathcal{R}_\zeta^* \right) ds. \quad (32)$$

It is positive. In fact, the supporting-line property of the convex set $\overline{\Omega}$ gives

$$\nu(\zeta) Z_\zeta^* + \overline{\nu(\zeta)} Z_\zeta = 2 \operatorname{Re}(\overline{\nu(\zeta)} Z_\zeta) \succeq 0. \quad (33)$$

Indeed, for every unit vector $x \in \mathbb{C}^n$, the quadratic form on the left is

$$2 \operatorname{Re}(\overline{\nu(\zeta)} (\zeta - x^* B x)) \geq 0,$$

because $x^* B x \in W(B) \subset \Omega$. Multiplication on the left by $\mathcal{R}_\zeta^*$ and on the right by $\mathcal{R}_\zeta$ turns the left side of (33) into the matrix in parentheses in (32).

Along the counterclockwise boundary, $d\zeta = i\nu(\zeta)ds$. The matrix Cauchy formula gives

$$\frac{1}{2\pi} \int_{\partial\Omega} \nu(\zeta) \mathcal{R}_\zeta ds = I. \quad (34)$$

Taking adjoints gives the same identity for the second layer. Hence

$$\int_{\partial\Omega} d\mu_B = 2I. \quad (35)$$

For $\varphi \in C(\partial\Omega)$, define

$$\Phi(\varphi) = \frac{1}{2} \int_{\partial\Omega} \varphi(\zeta) d\mu_B(\zeta). \quad (36)$$

By (35), Φ is unital and positive, hence $*$ -preserving; moreover, $\|\Phi(\varphi)\| \leq \|\varphi\|_\infty$, where

$$\|\varphi\|_\infty = \max_{\zeta \in \partial\Omega} |\varphi(\zeta)|.$$

For holomorphic h , the first layer of $2\Phi(h)$ is $h(B)$ by Cauchy's formula. The function g_h in (31) is holomorphic on Ω , so $g_h(B)$ is defined by the holomorphic functional calculus. To verify the relevant integral formula without assuming that g_h extends across $\partial\Omega$, choose a positively oriented contour $\Gamma \subset \Omega$ enclosing $\sigma(B)$. Cauchy's formula and Fubini's theorem give

$$\begin{aligned} g_h(B) &= \frac{1}{2\pi i} \int_{\Gamma} g_h(z)(zI - B)^{-1} dz \\ &= \frac{1}{2\pi i} \int_{\partial\Omega} \overline{h(\zeta)}(\zeta I - B)^{-1} d\zeta. \end{aligned} \quad (37)$$

Taking adjoints in (37) and using $d\bar{\zeta} = -i\overline{(\zeta)}ds$ gives the second layer of $2\Phi(h)$. This proves (30). $\square$

Proposition 4.2 (Cayley completion for an auxiliary pair). *Let B and Ω be as in Lemma 4.1, and let f be holomorphic on a neighborhood of $\overline{\Omega}$ and satisfy*

$$\max_{\zeta \in \overline{\Omega}} |f(\zeta)| \leq 1. \quad (38)$$

Put $T = f(B)$. Then there is an analytic $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ satisfying

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (39)$$

and

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (40)$$

Moreover, $\sigma(T) \subset \overline{\mathbb{D}}$.

Proof. By (38) and spectral mapping from $\sigma(B) \subset \Omega$, we have $\sigma(T) \subset \overline{\mathbb{D}}$. For $w \in \mathbb{D}$, define on the boundary

$$c_w(\zeta) = \frac{1 + wf(\zeta)}{1 - wf(\zeta)} \quad (41)$$

and put

$$H(w) = \Phi(c_w). \quad (42)$$

The map $w \mapsto c_w$ is analytic with values in $C(\partial\Omega)$, locally uniformly on $\mathbb{D}$. Hence H is analytic, and $H(0) = I$. Furthermore,

$$\operatorname{Re} c_w(\zeta) = \frac{1 - |wf(\zeta)|^2}{|1 - wf(\zeta)|^2} \geq 0.$$

The positivity and $*$ -preservation of Φ give $\operatorname{Re} H(w) = \Phi(\operatorname{Re} c_w) \succeq 0$.

The boundary expansion, convergent in $C(\partial\Omega)$ locally uniformly for $w \in \mathbb{D}$,

$$c_w = 1 + 2 \sum_{m=1}^{\infty} w^m f^m \quad (43)$$

and (30) yield, with norm convergence,

$$H(w) = I + \sum_{m=1}^{\infty} w^m (T^m + g_{f^m}(B)^*). \quad (44)$$

Since the spectral radius of wT is less than 1, the Neumann series for $(I - wT)^{-1}$ converges in norm. Subtracting it from (44) gives

$$H(w) - (I - wT)^{-1} = \sum_{m=1}^{\infty} w^m g_{f^m}(B)^*. \quad (45)$$

Each g_{f^m} is holomorphic on a neighborhood of $\sigma(B)$. Finite-dimensional holomorphic functional calculus, equivalently Hermite interpolation followed by Cayley–Hamilton, gives $g_{f^m}(B) \in \text{alg}(B)$. Hence

$$g_{f^m}(B)^* \in \text{alg}(B^*).$$

Every partial sum on the right of (45) therefore belongs to $\text{alg}(B^*)$. This algebra is finite-dimensional and hence norm closed, which proves (40). $\square$

Corollary 4.3 (Auxiliary sharp bound). *In Proposition 4.2, if B has simple spectrum, then $\|f(B)\| \leq 2$.*

Proof. Write

$$B = S \text{diag}(\beta_1, \dots, \beta_n) S^{-1}.$$

Then

$$T = f(B) = S \text{diag}(f(\beta_1), \dots, f(\beta_n)) S^{-1}.$$

Since $\sigma(B) \subset \Omega$, the bound on f gives $|f(\beta_i)| \leq 1$ for every i . Apply Theorem 3.1 to these simultaneous diagonalizations and to the function H constructed in Proposition 4.2. The values $f(\beta_i)$ may repeat. $\square$

5. Passage to arbitrary matrices and proof of the main theorem

We now remove the hypotheses of simple spectrum and boundary regularity by successive approximations.

Lemma 5.1 (Simple spectrum and numerical range). *Matrices with n distinct eigenvalues are dense in $\mathbb{M}_n(\mathbb{C})$. Moreover, for $A, B \in \mathbb{M}_n(\mathbb{C})$,*

$$W(B) \subset W(A) + \{z \in \mathbb{C} : |z| \leq \|B - A\|\}. \quad (46)$$

Proof. The discriminant of the characteristic polynomial is a nonzero polynomial in the matrix entries, as is seen at $\text{diag}(1, \dots, n)$. Its nonvanishing set is therefore dense in $\mathbb{M}_n(\mathbb{C})$, proving the first assertion. For a unit vector x ,

$$|x^*(B - A)x| \leq \|B - A\|,$$

which proves (46). $\square$

Lemma 5.2 (Outer approximation). *If $K \subset \mathbb{C}$ is nonempty, compact, and convex, then there are bounded convex domains Ω_j with continuously differentiable boundaries such that*

$$K \subset \Omega_j, \quad d_H(\overline{\Omega_j}, K) \longrightarrow 0, \quad (47)$$

where d_H denotes the Hausdorff distance between compact sets.

Proof. For $\delta > 0$, let

$$\Omega_\delta = \{z \in \mathbb{C} : \text{dist}(z, K) < \delta\}.$$

This is a bounded convex domain and its closure is the Minkowski sum $K + \delta\overline{\mathbb{D}}$, so its Hausdorff distance from K is δ . Identify $\mathbb{C}$ with $\mathbb{R}^2$. The metric projection $\pi_K: \mathbb{C} \rightarrow K$ is unique and continuous. Put $d_K(z) = \text{dist}(z, K)$. For $z, h \in \mathbb{C}$, comparison first with $\pi_K(z)$ and then with $\pi_K(z + h)$ gives

$$\begin{aligned} 2 \text{Re}(\overline{z + h - \pi_K(z + h)} h) - |h|^2 &\leq d_K(z + h)^2 - d_K(z)^2 \\ &\leq 2 \text{Re}(\overline{z - \pi_K(z)} h) + |h|^2. \end{aligned}$$

The continuity of π_K now shows that d_K^2 is continuously differentiable, with real gradient $2(z - \pi_K(z))$. On the level set $d_K(z) = \delta$ this gradient has norm 2δ and hence never vanishes. Since $\partial\Omega_\delta = \{d_K = \delta\}$,

the implicit-function theorem shows that $\partial\Omega_\delta$ is continuously differentiable. Taking $\Omega_j = \Omega_{1/j}$ for $j \geq 1$ proves the lemma. $\square$

Proof of Theorem 1.1. Let A and p be arbitrary as in Theorem 1.1. Choose a bounded convex domain Ω with continuously differentiable boundary such that

$$W(A) \subset \Omega. \quad (48)$$

We first prove

$$\|p(A)\| \leq 2 \max_{z \in \overline{\Omega}} |p(z)|. \quad (49)$$

Because $W(A)$ is compactly contained in Ω ,

$$\text{dist}(W(A), \mathbb{C} \setminus \Omega) > 0.$$

Lemma 5.1 and (46) therefore give a sequence of matrices $B_k \rightarrow A$ having simple spectrum and satisfying $W(B_k) \subset \Omega$. Since $p(B_k) \rightarrow p(A)$, it is enough to prove (49) for an arbitrary such matrix B .

Set

$$m_\Omega = \max_{z \in \overline{\Omega}} |p(z)|. \quad (50)$$

If $m_\Omega = 0$, the polynomial vanishes on the open set Ω and is identically zero. In the remaining case $m_\Omega > 0$, put

$$f = \frac{p}{m_\Omega}. \quad (51)$$

Then $\max_{z \in \overline{\Omega}} |f(z)| \leq 1$. Corollary 4.3, applied directly to B and f , gives

$$\|f(B)\| \leq 2.$$

Equivalently,

$$\|p(B)\| \leq 2m_\Omega. \quad (52)$$

Letting the matrices B_k with simple spectrum tend to A proves (49).

Apply Lemma 5.2 to the compact convex set $K = W(A)$, obtaining Ω_j . All these closures eventually lie in one fixed compact neighborhood of K . Uniform continuity of p there and (47) give

$$\max_{z \in \overline{\Omega_j}} |p(z)| \longrightarrow \max_{z \in W(A)} |p(z)|. \quad (53)$$

Apply (49) to each Ω_j and pass to the limit. This proves (3).

It remains to justify the spectral-set conclusion. Let r be rational with no pole on $W(A)$. Since $W(A)$ is compact and disjoint from the finite set of poles of r , choose $\delta > 0$ so that the closed δ -neighborhood K_δ of $W(A)$ contains no pole. Thus r is holomorphic on an open neighborhood of the compact convex set K_δ . Since $\mathbb{C} \setminus K_\delta$ is connected, polynomial Runge approximation gives polynomials p_j converging uniformly to r on K_δ . As $\sigma(A) \subset W(A) \subset \text{int } K_\delta$, choose a positively oriented contour $\Gamma \subset \text{int } K_\delta$ enclosing $\sigma(A)$. Uniform convergence on Γ , together with the Cauchy integral formula for the holomorphic functional calculus, gives $p_j(A) \rightarrow r(A)$. Also

$$\max_{z \in W(A)} |p_j(z)| \longrightarrow \max_{z \in W(A)} |r(z)|.$$

Apply the polynomial estimate to p_j and pass to the limit to obtain (2) with $K = W(A)$ and $C = 2$. $\square$

6. Consequences and scope

Theorem 1.1 immediately gives the corresponding estimate for functions holomorphic on a neighborhood of the numerical range.

Corollary 6.1. Let $A \in \mathbb{M}_n(\mathbb{C})$, and let f be holomorphic on a neighborhood of $W(A)$. Then

$$\|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|. \quad (54)$$

Proof. Choose $\delta > 0$ so that the closed δ -neighborhood of $W(A)$ lies in the open set on which f is holomorphic. Polynomial Runge approximation on this compact convex neighborhood, followed by the Cauchy integral formula for the holomorphic functional calculus, gives the assertion. $\square$

The Hilbert-space version follows by the standard finite-dimensional compression argument; compare [13].

Corollary 6.2 (Hilbert-space form). Let $\mathcal{H} \neq \{0\}$ be a complex Hilbert space, with inner product linear in the first variable, and let $\mathcal{B}(\mathcal{H})$ denote its algebra of bounded operators. For $A \in \mathcal{B}(\mathcal{H})$, put

$$W(A) = \{\langle Ax, x \rangle : x \in \mathcal{H}, \|x\| = 1\}.$$

Then, for every $p \in \mathbb{C}[z]$,

$$\|p(A)\| \leq 2 \sup_{z \in W(A)} |p(z)|. \quad (55)$$

Consequently, $\overline{W(A)}$ is a 2-spectral set for A .

Proof. The assertion is immediate if $p = 0$. Otherwise, write $d = \deg p$. For a unit vector $x \in \mathcal{H}$, let

$$\mathcal{M}_x = \text{span}\{x, Ax, \dots, A^d x\}, \quad A_x = \Pi_x A|_{\mathcal{M}_x},$$

where Π_x is the orthogonal projection onto $\mathcal{M}_x$. Induction gives

$$A_x^k x = A^k x \quad (0 \leq k \leq d), \quad (56)$$

because $A^{k+1}x \in \mathcal{M}_x$ whenever $k < d$. Moreover, $W(A_x) \subset W(A)$. Theorem 1.1, applied on the finite-dimensional space $\mathcal{M}_x$, therefore gives

$$\|p(A)x\|_{\mathcal{H}} = \|p(A_x)x\|_{\mathcal{H}} \leq \|p(A_x)\| \leq 2 \sup_{z \in W(A)} |p(z)|.$$

Taking the supremum over unit vectors x proves (55).

Put $K = \overline{W(A)}$. The standard inclusion $\sigma(A) \subset K$ shows that $r(A)$ is defined for every rational function r without poles on K . The polynomial-approximation argument in the proof of Theorem 1.1, now applied to K , gives polynomials p_j such that $p_j(A) \rightarrow r(A)$ and

$$\sup_{z \in W(A)} |p_j(z)| \longrightarrow \sup_{z \in W(A)} |r(z)|.$$

Passing to the limit in (55) proves the rational estimate and hence the spectral-set assertion. $\square$

The conclusion also controls polynomial and rational approximation of matrix functions. Let $A \in \mathbb{M}_n(\mathbb{C})$, let f be holomorphic on a neighborhood of $W(A)$, and let s be either a polynomial or a rational function with no pole on $W(A)$. Then $f - s$ is holomorphic on a neighborhood of $W(A)$, and Corollary 6.1 gives

$$\|f(A) - s(A)\| \leq 2 \max_{z \in W(A)} |f(z) - s(z)|. \quad (57)$$

This is the sharp universal form of the numerical-range estimate that underlies applications to GMRES and rational Krylov methods; see [7,25,26] for the role of spectral sets in those settings.

7. Discussion and further directions

Theorem 1.1 proves the original, scalar-valued Crouzeix conjecture. The most immediate strengthening is the uniform matrix-polynomial question: for every $m \geq 1$ and every matrix polynomial $\mathcal{F} = [f_{ij}] \in \mathbb{M}_m(\mathbb{C}[z])$, does one have

$$\|\mathcal{F}(A)\| \leq 2 \max_{z \in W(A)} \|\mathcal{F}(z)\|, \quad \mathcal{F}(A) = [f_{ij}(A)]. \quad (58)$$

Here $\|\mathcal{F}(A)\|$ is the operator norm on $(\mathbb{C}^n)^m$, whereas $\|\mathcal{F}(z)\|$ is the operator norm on $\mathbb{C}^m$. The case $m = 1$ is precisely the original conjecture proved above; uniformity in m is an additional operator-algebraic requirement. The foundational matrix-level disk estimate is due to Okubo and Ando [27]. Uniform matrix-level spectral-set bounds and their relation to numerical radius are studied in [28,29]; the constant $1 + \sqrt{2}$ at every matrix level follows from [4], while sharp estimates for ρ -contractions with $1 \leq \rho \leq 2$ are developed in [30]. The extension is known with constant 2 for weighted shift matrices [18]. For block-valued inputs, the coefficients in the completion need not reduce to a single diagonal algebra, so the cancellation in (19) would require a genuinely operator-algebraic formulation.

The positive-real completion principle is independent of numerical ranges: it starts with a Carathéodory function whose resolvent defect lies in the adjoint algebra of an auxiliary simple-spectrum matrix sharing an eigenbasis with the matrix in the resolvent. A direct infinite-dimensional version of Theorem 3.1, based for example on spectral measures or Riesz bases, could reveal equality cases or uniform matrix-level functional-calculus structure. The main obstacle is to replace the finite diagonal correction supplied by the auxiliary eigenbasis with a stable functional-calculus condition that does not rely on a finite diagonal spectral model. Related questions for unbounded numerical ranges are studied in [31], and related abstract spectral-constant problems in [32,33].

Double-layer methods beyond numerical ranges were developed in [7], while configuration-constant refinements were introduced in [23]; see also [22] for a comparison of double-layer approaches. Annular double-layer kernels are studied in [34], while [35] develops a complementary annular realization and norm theory; elliptic domains admit structured dilation models [36]. Whenever a companion transform lands in a controlled adjoint algebra, the kernel-sampling argument may replace a triangle-inequality estimate and thereby offer a route to sharp constants for other geometric spectral sets.

With the notation of the proof of Theorem 3.1, if $\|\tilde{T}\| = 2$ and $e \in \mathbb{C}^n$ satisfies $\|e\|_2 = 1$ and $\|\tilde{T}e\|_2 = 2$, then equality throughout the last operator-inequality chain in the proof of Theorem 3.1 forces

$$\hat{P}e = 2e, \quad \hat{P}\tilde{T}e = \tilde{T}e.$$

Analyzing these vector saturation conditions may classify extremal pairs (A, p) and quantify stability near the nilpotent example (4). Such a classification must allow the nonuniqueness phenomena found in [12].

Finally, O’Loughlin and Rani extend the Crouzeix–Palencia framework to scaled q -numerical ranges, $q \in [0, 1]$, and propose associated spectral-set problems [37]. These provide a natural testing ground for positive-real completion, although no constant-2 analogue is asserted here.

AI-assistance disclosure

OpenAI ChatGPT contributed the idea of sampling the matrix Herglotz kernel at half the conjugate eigenvalues and adding the origin sample that cancels the adjoint-algebra correction; this appears in the proof of Theorem 3.1, especially (17)–(19). It also assisted with exposition, bibliography, and typesetting. The human author assumes full responsibility for the correctness of all mathematical statements and for the integrity and accuracy of all citations.

Acknowledgments

The author thanks Elijah Winners for suggesting the auxiliary-basis simplification used in the proof of Theorem 3.1.

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